

SODIUM TECHNOLOGY AND EQUIPMENT OF THE BN-350 INSTALLATION*

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1. Introduction

The general parameters and information on the BN-350 installation to be built in the U.S.S.R. on the Mangyshlak peninsula were given at the 1964 Geneva Conference⁽¹⁾ and the 1965 Detroit Conference.⁽²⁾ The present paper is devoted to a detailed description of the main engineering equipment and the experimental work carried out in support of the selected designs, and to an explanation of certain problems of sodium technology. The design and experimental work on the main circulating pumps, the intermediate heat exchanger, and the steam generator are examined.

2. Sodium Circulating Pumps

In the development of the pumps for the BN-350 plant, pumps were chosen with a free sodium level in the so-called bracket design (without bearings operating in the coolant), with biological shielding and a mechanical seal, for which operating experience existed.

Table 1 gives the parameters of the primary and secondary circuit pumps.

There is no basic structural difference between the primary and secondary circuit pumps except that the secondary circuit pump has no biological shielding.

The pump shaft is mounted on one radial and one radial-thrust sliding bearing. The distance between the axes of the impeller and the lower bearing is 2 m. To reduce the radial forces on the bearings in the pump a guide system is used. The biological shield is arranged inside the pump housing and

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part of the shield is in the sodium. Between the cover of the pump housing and the lower bearing there is a cooling zone to reduce the axial flow of heat towards the bearings and to prevent sodium vapours entering the bearing cavity of the pump.

The region is cooled with a sodium-potassium alloy. When necessary, the general oil system can switch on a shaft cooling system. Appropriate measures are taken in the pump to prevent the penetration of oil vapours and the oil itself into the sodium. Between the upper bearing and the coupling connecting the shafts of the pump and the electric motor there is a mechanical seal. The pump has the oil, gas, and other systems characteristic of equipment

TABLE 1. DETAILS OF THE PRIMARY AND SECONDARY PUMPS

Value: Parameter	Capacity (m ³ /h)	Head of sodium (m)	Shaft speed (rev/min)	Maximum rating of electric motor (kW)
Primary circuit	3220	110	1000/250	1700
Secondary circuit	3850	70	1000/250	1100

of this type. The pump mechanism can be removed from the housing without cutting the sodium piping. The bearings and upper oil seal can be inspected and repaired without taking off the removable part of the pump. The gas cavity in this case is sealed by a special standing seal which operates when the pump is stopped. There are small coolant flows through the internal clearances of the pump (between the rotating and stationary parts, and the fixed and removable parts). Part of this flow is fed back, the rest is passed to the intake side of the pump via a special drain line. Although there is no provision for governing the speed of the pump, it can operate under different conditions. Five, four, and three pumps can be connected in parallel with the reactor. In addition, as indicated above, the pump can operate at quarter speed. The corresponding automatic equipment is therefore provided, which ensures that sodium flows from the housing to the intake of the pump without entrainment of gas.

To prove the chosen pump design, experimental work has been and is still going on.

The geometry of the flow passage and the guide apparatus was chosen and then, using a 1:4.5 scale model, the number and direction of the internal leaks were determined, cavitation tests were carried out, the magnitude and direction of the hydraulic forces were defined, and the efficiency of the pump (70%) was determined.

The pump uses a seal of the type used in hydrogen cooled generators. The frictional heat from the bearing pairs is transferred to the oil and is removed

by a specially fitted cooler. The design of the bearing pairs was developed and selection of materials made on a special rig. It was established that the different materials performed satisfactorily. At present, preference is given to a combination of graphite and chrome steel surface. The oil leakage is a few cm^3/day .

On a full scale model of the moving parts, with all the parts except the impeller, the weight of which was simulated by a metal disk, the bearing design was developed, the temperature conditions of the shaft were checked with both the cooling pump and shaft oil cooling system in operation and switched off. It was found that the axial flow of heat along the shaft is an insignificant amount and the shaft cooling systems can fully cope with it.

A sodium rig is now under construction for testing the pumps. The rig is equipped with all the systems characteristic of a sodium loop (indicating and cold trapping systems, heat removal system, etc.) and the necessary instrumentation.

The circuit of the rig contains about 20 m^3 of sodium, and is made of piping 500 mm in diameter as on the BN-350 installation. On the rig, apart from testing the pumps with all the systems in conditions similar to real conditions, tests will also be carried out on an electric motor with its systems and the fittings, non-return valve, level-meter and flowmeter. The main task of the rig, however, is the complete operational testing of the previously developed main items of the pump and the pumping unit as a whole.

The rigs for the primary and secondary circuit pumps are side by side. Therefore, by fitting a connexion between them, the intention is to test the non-return valve in conditions simulating stoppage of one of the pumps of BN-350.

3. Intermediate Heat Exchanger

The intermediate heat exchanger is a shell and tube structure in two sections connected in parallel. Each section is made in the form of a horizontal vessel of rectangular cross-section with three heat transfer bundles in it. Each bundle consists of 343 U-shaped tubes 28 mm in diameter and 2 mm wall thickness arranged with a pitch sideways and of 35 mm in depth. The heat exchanger is made of 18Cr9Ni steel. The bundles in each section are connected in series in relation to the coolant in the primary circuit and in the secondary circuit. Movement of the coolants is counter-current with cross-flow. The radioactive sodium of the primary circuit enters the space between the tubes where it gives up its heat to the sodium in the secondary circuit which is moving inside the tubes. The vertical U-shaped tube bundle has a cylindrical shape. Therefore, to ensure uniform distribution of the flow of

coolant across the shell, the tube system of each bundle is extended to a rectangular shape by fitting fixed non-operative straight tubes.

To create a uniform primary sodium flow an equalizing grid is fitted at the entry to the first bundle. When necessary, any bundle in the heat exchanger can be taken out and replaced by a new one. To ensure the possibility of carrying out the work of extraction of the bundle, biological shielding is provided at its inlet and outlet chamber in the secondary circuit, with special channels for the passage of the secondary sodium. The bundle is fastened to the neck with a special locking device and the bundle chamber and neck are sealed by moustache welding.

The flat side walls of the shell of each section of the heat exchanger have a stiffening frame and anchor stays.

The main characteristics of the heat exchanger are:

Thermal capacity	200×10^3 kW
Primary circuit sodium temperature:	
(a) at entry to heat exchanger	500°C
(b) at exit from heat exchanger	300°C
Secondary circuit sodium temperature:	
(a) at entry to heat exchanger	273°C
(b) at exit from heat exchanger	453°C
Heat transfer surface	1120 m ²
Hydraulic resistance of primary circuit	0.146 kg/cm ²
Hydraulic resistance of secondary circuit	2.24 kg/cm ²

To confirm the technical decisions on which the design of the intermediate heat exchanger is based, the following experimental work was carried out.

(i) The hydrodynamics of the space between the tubes of the heat exchanger was investigated using a 1:2.5 scale model. The entrance pipe of the model was connected to a vacuum chamber by a pipe containing a measuring nozzle to determine the total flow of air through the model. The local velocities at various cross-sections of the model were measured with Pitot tubes. It was found that the inlet of the model did not produce uniform flow around the first tube bundle. This was because of the high expansion ratio of the flow and the proximity of the inlet pipe to the bundle.

To equalize the flow before the first bundle, a grid was fitted, made of dummy tubes in a staggered arrangement similar to the arrangement of the heat transfer tubes. It was also noted that the flow around the tube bundle in the upper and lower part was uneven. In the upper part the unevenness was caused by the increase in the flow cross-section because of the neck in which the tube plate of the bundle is fitted. In the lower part the flow velocities increased in the direction of the bottom of the shell, owing to a gap between the bottom and the tube bundle, and by the fairly rarefied region of the tubes where the flow is longitudinal. The unevenness in the vertical flow around the bundles was eliminated by making appropriate alterations.

The total, experimentally determined, hydraulic resistance coefficient of the heat exchanger was 2.8 and agreed quite well with the theoretical value.

(ii) Vibration investigations of the heat exchanger tubes were carried out on a model with a transverse flow of coolant. The geometry of the model bundle, the method of securing, and the materials were similar to the authentic bundles. The experiments were carried out in water under conditions of hydrodynamic similarity. The model of the heat exchanger operated at cross-flow velocities in the range 0.4–0.7 m/sec for 1600 hours. No breaks were observed in the sealing of the tubes in the tube grids, nor was there any wear. The vibration frequency of the tubes at velocities of 0.4–0.865 m/sec was stable and amounted to 4–6 c/sec. The maximum amplitude of oscillation of the stresses in the tubes was recorded at the maximum velocity of 0.865 m/sec and amounted to 25 kg/cm². The experiments showed that the dynamic stresses arising in the tubes under the effect of a transverse flow of coolant were small for this design and could not greatly affect the performance of the heat exchanger.

In addition to the model studies it is intended to carry out vibration tests on a full scale heat exchanger, circulating water through the primary and secondary circuits at the nominal flow velocities.

(iii) A sodium rig with a thermal capacity of 3000 kW has been built and is in operation for confirming experimentally the thermal characteristics of the heat exchanger.

The experiments will make it possible to determine:

- (a) the adequacy of the heating surface and that the methods of calculation used in the design are correct;
- (b) the temperature distribution along the heat transfer surface. The presence and the effect of convective sodium currents in the spaces between the tubes;
- (c) the performance of the heat exchanger in long term operation under normal conditions and under thermal shock conditions.

4. Steam Generator

The steam generator consists of two sections, connected in parallel in the sodium and steam/water circuits, a gas chamber and connecting piping.

Each section of the steam generator includes one steam superheater and one evaporator. The evaporator is a vertical vessel with natural circulation of water inside field tubes and with steam separation in the steam chamber of the vessel.

Sodium enters the evaporator from the superheater and moves upwards through the space between the tubes, giving up its heat to the water. Water from the evaporator enters the internal down channel of the field tube,

moves downwards, enters the annular gap where it is heated by the sodium and, partially evaporating, moves upwards. The steam-water mixture forming is carried through the steam discharge risers into the steam chamber of the evaporator. The water level in the evaporator is 150 mm below the tops of the steam discharge risers.

The tube plate divides the evaporator vessel into two, the top part being the steam-water chamber and the lower part, the sodium chamber.

In the upper part are the feedwater supply header, the blow-through header, the emergency dump line for the water from the evaporator, a jet extinguishing plate with down tubes, and separation devices.

The lower sodium chamber can be divided into three: inlet chamber, working section, and outlet chamber with gas chamber.

In the inlet chamber there is a double perforated plate to equalize the flow of sodium over the cross-section of the evaporator.

The working section consists of a vessel 1400 mm in diameter and 24 mm wall thickness, in which there are 816 field tubes in a triangular lattice with a pitch of 44 mm. The field tube consists of an outer tube with a diameter of 32 mm and 2 mm wall thickness and an inner tube, 16×1.4 mm. The lower end of the 32×2 mm tube is rolled and sealed. The top end of the 16×1.4 mm tube is fastened to a steam discharge riser 600 mm high fitted in a socket of the tube plate.

At the top of the outlet chamber there is a gas chamber for removal of the gaseous reaction products of the interaction of sodium and water in the event of bursts in the tube system. The sodium level in the outlet chamber is below the tube plate and intersects the field tubes. This position of the level need not cause misgivings about the reliability of the design since the temperature difference of the sodium and steam-water mixture in the region of this level does not exceed $3-5^{\circ}\text{C}$.

The steam superheater is a vertical U-shaped heat exchanger with the sodium and steam inlet and outlet chambers at the top. The heat transfer surface is made up of 805 U-shaped tubes 16×2 mm. The ends of the tubes are sealed into tube plates with the orifices arranged in a triangular pattern with a pitch of 23 mm.

The evaporator and steam superheater components are made of 2CrMo steel. The gas chamber is a cylindrical vessel with spherical ends. It is intended for partial trapping of the sodium which may be ejected with hydrogen from the steam generator in the case of a substantial burst in the heating surface, and for compensating the volume expansion of the sodium in the secondary circuit. The interconnecting pipe system consists of sodium, steam-water lines and piping for emergency discharge of sodium-water reaction products.

The sodium, coming from the heat exchanger along a tube 500 mm in diameter, is passed via two tubes, 350 mm in diameter, to the steam superheaters. From the steam superheaters the sodium enters the evaporators via

a similar system and continues along a tube, 500 mm in diameter, to the secondary circuit pump.

Feedwater is passed to the evaporators along tubes 150 mm in diameter. On each evaporator line there is a feed regulator and a quick acting, non-return stop valve. Variation of the water level in the evaporator causes a signal to be sent to the feed regulator. The quick acting, cut-off valve operates if a leak occurs in the steam generator.

For rapid elimination of trouble in the event of large leakage of water into the sodium, there is an emergency dump line for the water from the evaporators.

To discharge the reaction products there are tubes, 350 mm in diameter, connecting the gas cavities of the evaporators with the gas chamber. The latter is connected by a system of tubes, including a safety device, to the reaction product separator which is fitted inside the steam generator building.

For removing from the steam superheaters the gaseous products which may be given off under the tube plates in the event of slight leaks of water into the sodium, the top of each superheater vessel is connected to the gas cavity of the evaporators by tubes, through which a slight constant sodium flow is maintained under normal operating conditions.

Main characteristics of the steam generator:

Steam output	276 t/h
Superheated steam pressure	50 kg/cm ²
Pressure in the evaporator	53 kg/cm ²
Superheated steam temperature	435°C
Feedwater temperature	158°C
Sodium temperature at entry to steam superheater	453°C
Sodium temperature at exit from steam superheater	416°C
Sodium temperature at exit from evaporator	273°C
Steam generator resistance with respect to steam	1.5 kg/cm ²
Steam generator resistance with respect to sodium	0.93 kg/cm ²
Including:	
Evaporator	0.32 kg/cm ²
Steam superheater	0.47 kg/cm ²
Steam superheater/evaporator piping	0.14 kg/cm ²
Heating surface of evaporator	820 m ²
Heating surface of steam superheater	455 m ²
Number of passes through the evaporator	4

Up to the present time the following experimental work has been carried out in support of the selected design:

(i) Three series of experiments have been carried out to investigate the effects of interaction of sodium and water on fracture of a heat transfer tube.

In the first two series a study was made of the influence of the composition

of the gas cushion on the interaction, the influence of the speed and direction of the sodium flow on the variation of temperature at the point of contact between the reagents, and on the propagation of gaseous reaction products. Water or steam was fed into the sodium via a special device—a needle valve. Since, in the BN-350, the possibility of a large heat transfer tube suddenly opening up is not excluded, in the last series of experiments a study was made of the stability of a tube bundle in the evaporator in the event of a large hole in one of the tubes with simultaneous entry of a large quantity of water into the sodium. The experiments were carried out on a model which was made from the same materials adopted in the project. The experimental rig is a closed sodium circulation circuit including a model of an evaporator, an electromagnetic pump, a transport and drainage vessel, fittings, and piping.

The shell of the model was made of two tubes, 245 mm in diameter and 13 mm wall thickness and 426 mm in diameter and 16 mm wall thickness. In the top part of the space between the tubes there was a gas cavity left after filling with sodium. Inside the shell was a tube bundle made of 2CrMo steel 32 mm in diameter; eighteen tubes in the bundle were fixed, and one, in the centre, was removable. All the tubes were blocked at the bottom. The lattice pitch of the tubes and the spacing were exactly the same as in the real evaporator of the BN-350 plant. The bundle was sealed to the shell by moustache welding and fastened by studs. The cover of the model was sealed on the steam–water chamber with a paranite gasket.

The central removable tube was intended to feed water into the sodium in the space between the tubes and had a weakened ring shaped part at a certain point designed to give a large burst at pressures of 50–60 kg/cm².

The weakness was an annular reduction in the thickness of the tube to 0.5 mm over a length of 50 mm and a further reduction to 0.1 mm over the same length by means of a longitudinal bald spot.

The batcher fitted on the flange of the central tube was in the form of a cylinder with a piston, to the bottom of which was fastened a rod designed to cover the conical orifice of the central tube after the water injection was completed.

To measure the pressure pulsations at the point of fracture and in the gas chamber of the model, pressure gauges were used together with the corresponding amplification and supply systems and a loop oscillograph. Recordings of the variation of the sodium temperature at the point of contact and up the space between the tubes in the section were achieved with specially designed low inertia, chromel–alnmel microthermocouples.

One of the most important measurements noted was the recording of the variation with time of the position of the piston during the injection of water. Altogether five experiments were carried out with up to 3 kg of water entering the sodium (the sodium capacity of the rig was 100 litres). The injection time

varied within the limits of 3–4 sec. The results in all the experiments were approximately the same.

At the injection point the pressure rose instantaneously at the moment of fracture to 40–50 kg/cm², then there was some pulsation and a drop in this pressure. The flow of sodium through the model practically stopped, the temperature at the point of injection rose briefly to 700–800°C. The pressure in the gas cavity first rose and then, after rupture of the bursting disk, fell, and its variation was determined by the carrying capacity of the reaction product dump system. In all the experiments it was found that almost all the sodium between the injection point and the gas cushion flowed into the dump system. There was never any fracture of the adjacent tubes filled with water at a pressure of 50–60 kg/cm². The bending of the central and adjacent tubes was slight.

(ii) The circulation characteristics of a field tube were investigated in a wide range of heat loads and water–chemical tests of the evaporator elements were carried out.

For these purposes a special rig was built with a single field channel. The number of passes was related to the water flow cross-sections in the down tube and annular slot for the value $F_{\text{tube}}/F_{\text{total}} = 0.111, 0.296, 0.814$. It was found that the highest number of passes was obtained for an area ratio in the range 0.3–0.4, irrespective of the magnitude of the heat flux which varied from 60×10^3 kcal/m²h to 160×10^3 kcal/m²h.

The geometry chosen for the field tube was checked for stable operation under various conditions. With variation of the heat load of the element in the range 60×10^3 to 260×10^3 kcal/m²h, variation in the feedwater temperature within the range 50–200°C and pressure fluctuation in the range 30–52 kg/cm², the number of passes varied within the limits of 2–10 with stable operating conditions maintained.

(iii) A sodium rig with a thermal capacity of 3000 kW was built and operated to check experimentally the thermal characteristics on models of the evaporator and steam superheater. The models were full size in length and had a number of heat transfer channels corresponding to the capacity of the rig.

The results of the experimental investigations to date are:

- (a) satisfactory agreement between the experimental and theoretical data on the heat transfer surfaces and temperature distribution;
- (b) stable hydrodynamics in the field tubes during operation under different conditions;
- (c) reliable operation of the separating devices of the evaporator model with regard to delivery of steam of the necessary quality for the alternative outlet for the steam–water mixture; viz. below the water level and directly into the steam volume;

- (d) satisfactory operation of the individual parts of the structure (fixing of tubes into tube plates, seals, welded joints, etc.). Up to the present the models have been in operation for more than 2000 hours. The experimental investigations are continuing.

5. Problems of Sodium Technology and Auxiliary Systems

As has already been mentioned, heat removal from the BN-350 reactor is achieved by means of six parallel systems, of which five are permanently in operation and one is in reserve. Each system of the primary circuit has independent connexions to the reactor and can be disconnected from the rest of the system by slide valves. The valves, with an elastic wedge, have two working surfaces and the cavity between them can be drained, which guarantees against leakage of sodium into the isolated loop.

Shut-off devices were fitted for two main reasons:

- (i) the presence of the slide valves makes it possible to isolate and drain the loop in the event of any faults in it;
- (ii) the need for enclosing the whole of the primary circuit in a containment vessel is eliminated for the case of fracture of the walls of equipment or piping.

In BN-350 only the reactor vessel and some of the piping (up to the slide valves) is enclosed in a shell filled with inert gas. The volume of the cavity between the vessel and the shell is such that filling it with sodium does not lead to interruption of the flow of circulating sodium and overheating of the core.

Sealing in relation to the rod of the main slide valves is achieved with a frozen sodium seal, cooled by a sodium-potassium eutectic. On the rod there is also a supplementary stuffing box of the ordinary type.

A characteristic feature of the valve design is the facility for remote extraction of its operating element for repair or replacement. For this operation it is not necessary to cut the piping of the primary circuit or enter the circuit building; it can even be carried out without draining the sodium from the circuits, since both slide valves are located approximately at the level of the top of the reactor. Access to the dismountable joints is assured by the built-in biological shield, as in the pump or intermediate heat exchanger.

The efficiency of the plant as a whole is guaranteed by a whole series of auxiliary systems. The most important auxiliary liquid metal systems include:

- (1) primary and secondary sodium purification system and indication of oxide content;
- (2) sodium-potassium systems;
- (3) primary and secondary circuit drain tanks;
- (4) coolant preparation system.

The purification and oxide indication systems are used to determine the quantitative sodium oxide content in the coolant of the primary and secondary circuits and to purify the latter.

In the primary circuit there are six cold traps. The number of traps operating will depend on the degree of contamination of the sodium. As a rule, one or two traps will be on at the same time, but their cooling system is designed for four. In addition there are two plugging indicators in which the sodium is cooled by air. Each loop of the secondary circuit has its own independent system including a cold trap and indicator.

Each cold trap is designed for a sodium flow of $10 \text{ m}^3/\text{h}$. Traps for the primary and secondary circuit are identical in design and consist of a vertical cylindrical vessel about 6 m high and 1 m in diameter. They are fitted in special electrical furnaces and all connexions to them are from above. The traps of the primary circuit are placed with the furnaces in a concrete shield. At the top they are closed off by a layer of cast iron providing access to the tubes when replacement is necessary.

Each trap has a built-in regenerator section, settling tank, and filter filling of stainless shavings. The temperature of the contaminated sodium at exit from the regenerator is always kept above the plugging temperature by passing the cold (clean) sodium through a special by-pass.

Cooling of the traps is effected by a eutectic alloy of sodium and potassium (78% K). The latter is passed through a coil in the settling tank and around the jacket surrounding it. The rates of flow through these zones can be adjusted individually, which fixes the necessary temperature conditions in the settling tank and the filter layer. The alloy temperature at entry to the trap is maintained at $50\text{--}60^\circ\text{C}$.

A special sodium-potassium system for heat removal from the cold traps includes two electromagnetic pumps with a capacity of $750 \text{ m}^3/\text{h}$ each and two air heat-exchangers. There is no standby in this system and in the event of any component of the system being out of order, some of the traps will be shut off.

A similar but independent sodium-potassium system is provided for cooling the seals of the fittings and pumps. Here there is a 100% standby system in view of the importance of uninterrupted cooling.

The drainage tanks of the primary circuit are designed to be able to drain the sodium from the whole circuit, and ten tanks each of 50 m^3 are provided. The tanks are connected in pairs to hold gas and metal respectively, and each pair of tanks is arranged in a separate box, which makes it possible to carry out the necessary servicing during operation. In addition, in an individual box there are two such tanks permanently connected to the gas cavity of the reactor. With these the fluctuations in gas pressure in the reactor on changes in the volume or temperature of the sodium are kept within admissible limits.

The drainage tanks of the secondary circuit are designed for simultaneous drainage of any two loops of the six (four tanks of 50 m³ each). The primary and secondary circuits are filled from the drainage tanks by means of linear electromagnetic pumps. Drainage of the sodium from the circuits is by gravity.

Apart from the filling operation, the pumps of the tank system of the primary circuit are used for purifying the metal in the drain tanks by pumping it through the cold traps.

In the BN-350 installation, measures are provided for repair and replacement of the sodium equipment of the main circuits. The removable parts of the pumps, heat-exchanger bundles, operating elements of the main slide valves are removed from the circuit in sealed pressurized bags in which an inert atmosphere is maintained. The bags permit the replacement of equipment without draining the sodium from the circuit. A special lock prevents contact between the inside of the circuit and the atmosphere. The equipment is transferred in these bags to a washing system. In this system the sodium residue on the components taken out of the circuit can be cleaned off by steam and water and where necessary they can be treated with deactivating solutions. After washing, the equipment is passed to the repair section.

References

1. A. I. LEIPUNSKII, O. D. KAZACHKOVSKII, I. I. AFRIKANTOV, M. S. PINKHASIK, N. V. KRASNOYAROV and M. S. POIDO, Sodium-cooled fast reactors. Paper 312, *3rd Geneva Conference*.
2. A. I. LEIPUNSKII *et al.*, The BN-350 atomic station. Paper at the *Detroit Conference*, April 26-28, 1965.